ParticularMatter Track Map

User guide: map the air you breathed along your route

19 October 2026

Getting started

What the app does

The app combines two files you already have:

- a **GPS track** (`.gpx`, e.g. exported from Strava): where you were;
- an **air-quality sensor log** (`.csv`, e.g. from a ParticularMatter / KongQi node): what the sensor measured.

It matches them **by time** and shows:

1. a **map of your route coloured by pollution level**;
2. a **time series** of the readings;
3. a **CSV download** with position + measurements for every reading.

Step 0: Install and start the app

Install R ≥ 4.1 and, once, the packages:

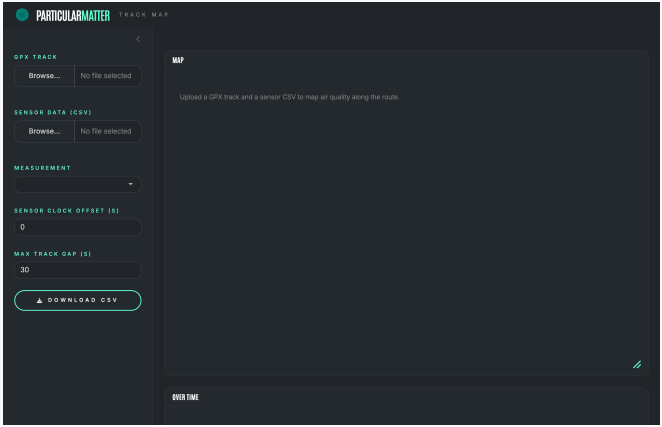
```
install.packages(c("shiny", "bslib", "leaflet",
                   "plotly", "xml2", "readr",
                   "dplyr", "lubridate", "tibble"))
```

Start the app in one of these ways:

- **RStudio:** open `particularmatter-gpx.Rproj`, open `app.R`, click **Run App**;
- **R console:** `shiny::runApp()`;
- **Terminal:** `Rscript -e 'shiny::runApp(launch.browser = TRUE)'`.

The app opens in your browser. An internet connection is needed for map tiles and fonts.

The app when it opens

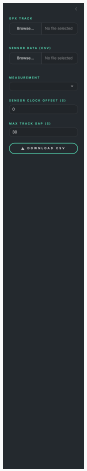


- **Sidebar (left):** files and settings.
- **Map** and **Over time** cards (right) fill in once files are loaded.
- Until then the map shows a hint: Upload a GPX track and a sensor CSV…

To try the app without your own data, use the demo files in `example/`.

Step by step

Step 1: Upload the GPX track

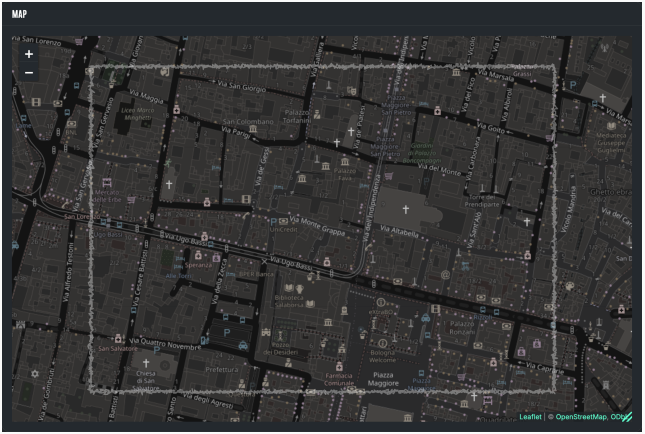


Click **Browse**… under **GPX track** and choose your `.gpx` file.

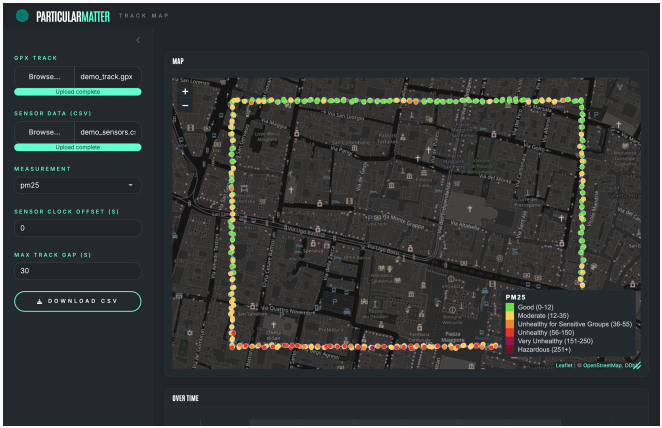
From Strava: open the activity on the website, click the **three dots** menu, then **Export GPX**. Any GPX file with timestamps works (Garmin, Komoot, phone apps…).

Demo: `example/demo_track.gpx`.

Step 1: The track appears on the map


MAP
Via San Lorenzo
Via San Giorgio
Via Magna
Via Farini
Via Monte Grappa
Via Ugo Bassi
Via Quattro Novembre
Via Altabella
Via Marsala
Via Goito
Via del Monte
Via Rizzoli
Via Indipendenza
Via Clavature
Via Orefici
Via Castiglione
Via Manzoni
Via Galliera
Via Cesare Battisti
Via Alberto Tristan
Via Oberdan
Via Regno d'Italia
Via Belle Arti
Via Zamboni
Via dei Mille
Via Guglielmo Oberdan
Via Augusto Righi
Via Giuseppe Garibaldi
Via Ugo Foscolo
Via Massimo d'Azeglio
Via Cesare Battisti
Via Riva di Reno
Via Santo Stefano
Via Castiglione
Via Nosadella
Via Mentana
Via Matteotti
Via Giuseppe Guglielmo
Via Ugo Bassi
Via Rizzoli
Via Indipendenza

Step 2: Upload the sensor CSV

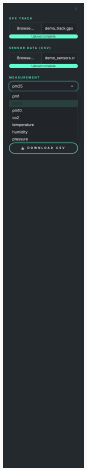


Click **Browse**… under **Sensor data (CSV)**. For a KongQi node, this is the log file copied from the SD card.

Demo: `example/demo_sensors.csv`.

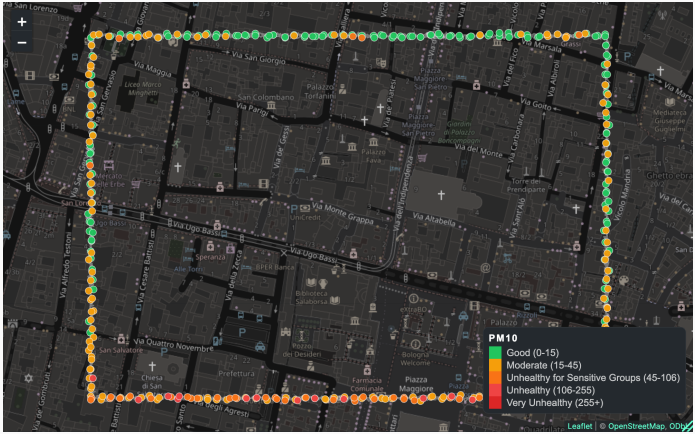
Each dot is now **one sensor reading**, placed where you were at that moment and **coloured by its air-quality band**.

Step 3: Choose a measurement



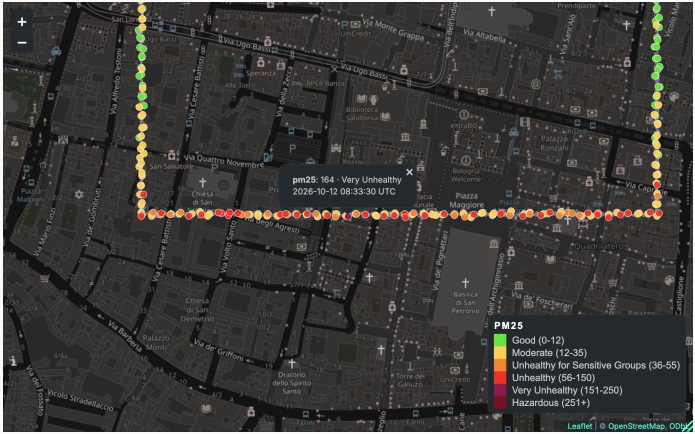
Open **Measurement** and pick what to map: `pm1`, `pm25`, `pm10`, `co2`, `temperature`, `humidity`, `pressure`, … Any numeric column in your CSV shows up in this list. PM2.5 is selected by default when it is present. Map, legend and chart update immediately.

Step 3: The same route, PM10



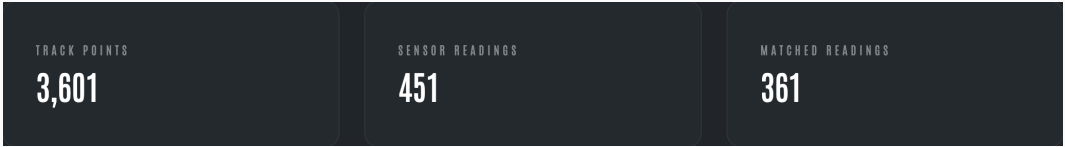
Changing the measurement re-colours the dots and the legend (here PM10, with its own bands).

Step 4: Read the map



Click a dot to see its value, band and time. Hover to see just the value. Use + / − or the mouse wheel to zoom.

Step 4: How many readings were matched?

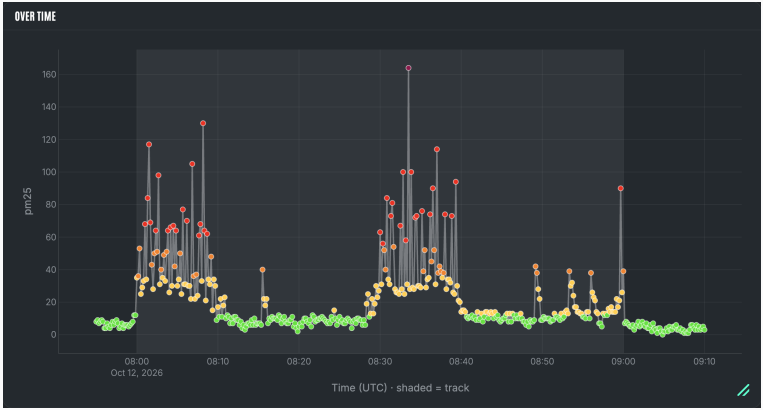


Below the chart, three boxes summarise the data:

- **Track points:** points in the GPX file;
- **Sensor readings:** rows in the CSV;
- **Matched readings:** readings that fell **inside the track's time range** and were placed on the map.

Readings taken before the track starts or after it ends are not mapped (here, the sensor ran 5 min before and 10 min after the walk).

Step 5: Check the time series



All sensor readings over time, in the same colours as the map. The lighter shaded area is the period covered by the GPS track. Hover for exact values.

Step 6: Download the combined data



Click **Download CSV**. You get `<track name>_air_quality.csv` with one row per matched reading:

Column	Meaning
`time`	Reading time (UTC, offset applied)
`lat`, `lon`, `ele`	Position from the GPX track
`pm25`, `co2`, …	All measurements
`time_source`	`gps`, `wifi` or `ts_ms`

Fine-tuning

Sensor clock offset

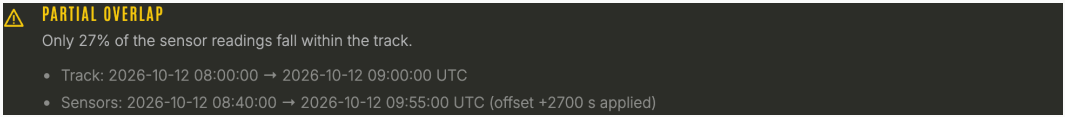
If the sensor clock was wrong by a known amount, enter the correction in **Sensor clock offset (s)**:

- a **positive** value moves the sensor data **later**;
- a **negative** value moves it **earlier**.

Example: the sensor clock was 2 minutes slow → enter `120`.

Map, chart and summary update immediately. Use the warnings on the next slides to check the result.

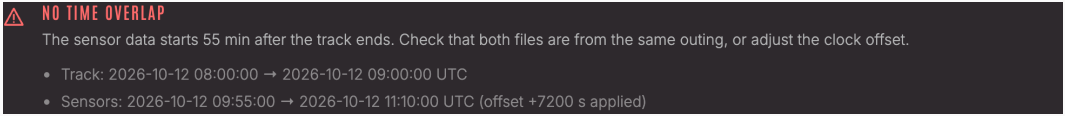
Warning: partial overlap



A **yellow** banner appears above the map when **fewer than half** of the sensor readings fall within the track. It lists both time ranges so you can compare them.

What to do: check that the sensor clock was set correctly, and correct it with the clock offset if needed.

Warning: no time overlap



A **red** banner means the two files **share no time at all**. The map shows only the grey track. The banner says how far apart the two files are.

What to do: make sure both files are from the **same outing**. If they are, the sensor clock is wrong: set the clock offset.

Max track gap and resizing

Max track gap (s), default 30 s. If the GPS recording paused (e.g. a stop at a café), readings taken during the gap are **dropped** rather than placed on a straight line between the two ends of the gap. Increase it if your GPS records points only rarely.

Resize the map or chart.

- Drag the **teal handle** in the bottom-right corner of a card.
- Or use the **full-screen** button in the card's corner.

Reference

Colour bands

Level	PM2.5 / PM1 (µg/m^3)	PM10 (µg/m^3)
Good	0 – 12	0 – 15
Moderate	12.1 – 35.4	15.1 – 45
Unhealthy for Sensitive Groups	35.5 – 55.4	45.1 – 106
Unhealthy	55.5 – 150.4	106.1 – 255
Very Unhealthy	150.5 – 250.4	255.1 +
Hazardous	250.5 +	–

- **CO_2 (ppm):** Good ≤ 800 · Moderate ≤ 1000 · Poor ≤ 1500 · Very Poor > 1500
- **Temperature, humidity, pressure:** comfort bands, shown in the legend
- **Any other column:** a teal scale from dim (low) to bright (high)

Sensor CSV format

One row per reading, with a header row. All times are in **UTC**.

Column	Needed?	Meaning
`gps_datetime`	one of the two	GPS time, e.g. `2026-10-12T08:05:10Z`
`wifi_datetime`	one of the two	Internet time, used if GPS time is missing
`ts_ms`	optional	ms since start; fills in missing times
`pm25`, `co2`, …	at least one	Measurements (any numeric column)
`lat`, `lng`, …	optional	Ignored: the GPX gives the position

Troubleshooting

Problem	What to do
Red **No time overlap** banner	Use files from the same outing, or set the clock offset
Only a grey line, no dots	No readings in the track's time range; read the banner
Map tiles or fonts missing	Connect to the internet
"Could not read GPX"	The GPX needs `<trkpt>` points with `<time>`; re-export it
"Could not read sensor CSV"	Add a `gps_datetime` and/or `wifi_datetime` column
Upload rejected as too large	The limit is 100 MB; change `shiny.maxRequestSize` in `app.R`
`there is no package called …`	Install the missing package

Summary

1. **Start** the app: `shiny::runApp()`.
2. **Upload** the GPX track.
3. **Upload** the sensor CSV.
4. **Choose** a measurement.
5. **Read** the map: click dots, check the matched count.
6. **Check** the time series and any overlap warning; fix the clock offset if needed.
7. **Download** the combined CSV.